Final Year Project Summary

Name: NURNISRIZA BINTI DE AFENDI
Matric Number: BI23110047
Program: UH6481005 Data Science

Project Title: Web-Based Public Transport Ridership Prediction and Travel Planning System

Supervisor: Dr. Ervin Gubin Moung

Project Type: Product Development

Problem Statement:

Major public transport systems, particularly in Malaysia's ETS services may experience fluctuations in passenger volumes which can lead to congestion during peak hours and reduced numbers during off-peak hours. Many travelers, particularly students, do not understand peak travel times and may be oversold on tickets or underestimate how much time they will need on the bus to get to their destination. Current systems are confined to ticketing and scheduling without the ability to intelligently forecast and provide real-time guidance on demand. Historical ridership data needs to be combined with user generated information from live mobile sensors and holiday or event awareness to facilitate improved planning for passengers, operators and administrators.

Objective:

1. To study the ETS ridership data in history to determine the trends for each route, at each time and season.
2. Building machine learning prediction approach based on historical data and live data obtained from the users' behavior.
3. To determine ridership demand levels (Low / Medium / High / Very High) and to reveal the peak and off-peak travel patterns.
4. To create a multi-role web-based application with dashboards for ETS Operators, Passengers and Administrator.
5. To incorporate holiday or event calendar awareness and multilingual support to improve the accuracy of predictions and make predictions more accessible to users.

Scope:

Product Modules:

1. Route Search & Selection - Create new route search query, read available ETS routes between origin & destination, update search parameters, delete search history

2. Ridership Prediction - Create new prediction for a route/date/time, read prediction results & demand level, update with different inputs, delete saved prediction records

3. Demand Classification - Read current demand classification for any route

4. Travel Recommendation - Read recommended departure times & alternatives, update recommendation for different dates, remove saved recommendations

5. Crowd Feedback Submission - Create crowd level feedback after travel, read submitted feedback history, update feedback within 24 hours, delete submitted feedback

6. Saved Trip Plans - Create bookmark / save a planned trip, read all saved trips, edit saved trip details, remove a saved trip plan

7. Trend Analysis & Visualization - Generate trend report for a specific period, read peak hour charts, busy routes, update data filters, remove custom trend reports.

8. Operator Analytics Dashboard - Create route analytics report, read ridership analytics, peak demand periods, update analytics by route, date, period, delete old generated reports

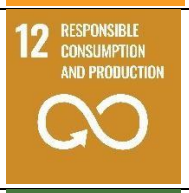
Supervisor Stamp of approval

Date:

Remarks (for Review Committee use)

https://sdgs.un.org/goals

Your project should aim to fulfil one of the 17 Sustainable Development Goals (SDG). Please map your project to one of the relevant SDG by ticking the relevant box.

SDG	Description	/	SDG	Description	/
	End poverty in all its forms everywhere			Reduce inequality within and among countries	
	End hunger, achieve food security and improved nutrition and promote sustainable agriculture			Make cities and human settlements inclusive, safe, resilient and sustainable	/
	Ensure healthy lives and promote well-being for all at all ages			Ensure sustainable consumption and production patterns	
	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all			Take urgent action to combat climate change and its impacts	
	Achieve gender equality and empower all women and girls			Conserve and sustainably use the oceans, seas and marine resources for sustainable development	
	Ensure availability and sustainable management of water and sanitation for all			Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss	
	Ensure access to affordable, reliable, sustainable and modern energy for all			Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels	
	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all			Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development	
	Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation				